Graduation Project Proposal

Project Title

AI-Based Smart Load Shedding and Energy Management System for Developing Countries

Project Team

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Project Overview

Electricity shortages are a major challenge in many developing countries. When the available power supply is lower than demand, electricity companies use load shedding to reduce pressure on the grid. However, traditional load-shedding methods often depend on fixed schedules or manual decisions, which may not reflect real-time needs or the importance of different services.

This project proposes an AI-based smart load-shedding and energy management system. The system will use simulation to study how limited electricity can be distributed more effectively during shortage periods. Different artificial intelligence methods may be used, including prediction models, rule-based decision support, optimization techniques, and reinforcement learning for selected simulation scenarios.

The project will focus on Sudan as a case study, but the general idea can be applied to other developing countries facing similar energy problems.

Problem Statement

Sudan faces repeated electricity shortages because power generation is often not enough to meet total demand. Current load-shedding approaches can be inefficient because they may disconnect areas based on fixed schedules without considering real-time demand, critical services, or fairness between communities.

This can negatively affect hospitals, water pumping stations, bakeries, schools, homes, and

small businesses. In such cases, the main problem is not only the shortage of electricity, but also the way the available electricity is managed and distributed.

There is a need for a smarter decision-support system that can help prioritize essential services, reduce unnecessary harm, and make load-shedding decisions more balanced and data-driven.

Research Questions

1. How can artificial intelligence be used to support smarter load-shedding and energy management decisions?
2. How can the system prioritize critical services such as hospitals, water stations, and bakeries during electricity shortages?
3. What types of data and rules are needed to simulate load-shedding decisions in a realistic way?
4. Can AI-based decision-making reduce unfair blackout durations between different areas?
5. How does the proposed system perform compared with traditional fixed load-shedding schedules?

Project Objectives

1. To study the problem of load shedding and electricity shortage in developing countries, with Sudan as the main case study.
2. To design a software simulation that represents electricity supply, demand, shortage levels, and different consumer sectors.
3. To develop an AI-based decision-support model for managing electricity distribution during shortage periods.
4. To include priority levels for critical infrastructure such as hospitals, water facilities, bakeries, and other essential services.
5. To test different AI methods, including prediction, optimization, rule-based logic, and reinforcement learning where appropriate.
6. To compare the proposed smart system with a traditional fixed-schedule load-shedding approach.
7. To evaluate the system using measures such as critical-service downtime, fairness between areas, and overall distribution efficiency.

Proposed Scope

The project will be implemented as a software-based simulation, not as a direct control system for a real national power grid. The simulation will model a simplified power distribution scenario where demand is higher than available supply. The AI component will recommend how power should be distributed or reduced based on priorities, shortage level, and fairness rules.

The project does not aim to solve the full national electricity crisis. Instead, it aims to show how AI can improve decision-making during load shedding and provide a practical foundation for future research.

Expected Deliverables

- A documented software simulation of a simplified power grid under electricity shortage conditions.
- An AI-based decision-support model for smart load-shedding and energy management.
- Simulation scenarios showing how the system handles different shortage levels and priority needs.
- A comparison between the proposed AI-based system and traditional fixed load-shedding schedules.
- Evaluation results showing critical-service downtime, fairness between areas, and overall power-distribution performance.
- A final graduation project report.
- A project presentation suitable for academic discussion.